

● Variables (Q1–Q6)

Q1. Declare variables to store your name, age, CGPA, and whether you are a student. Print all values.

Q2. Swap the values of two integer variables $a = 5$ and $b = 10$ without using a third variable.

Q3. Declare a variable $\text{price} = 100$. Apply a 20% discount and print the final price.

Q4. A variable $\text{distance} = 150$ (km) and $\text{time} = 3$ (hours). Calculate and print the speed.

Q5. Declare $\text{principal} = 5000$, $\text{rate} = 8$, $\text{years} = 3$. Calculate and print Simple Interest using $\text{SI} = (P * R * T) / 100$.

Q6. Declare two variables with the same value, then change one of them and print both to show variables are independent copies.

● Data Types (Q7–Q12)

Q7. Declare one variable of each primitive type (byte, short, int, long, float, double, char, boolean) and print all of them.

Q8. Take a double value like 9.99999 and store it in a float. Print both and observe the difference in precision.

Q9. Store the ASCII value of character 'Z' in an int variable and print it. Then cast it back to char and print again.

Q10. Create a String variable for a full name and a char variable for a grade. Print a sentence combining both: "Alice scored grade A".

Q11. Declare $\text{long population} = 8100000000\text{L}$. Print it, then divide by 1000000 to show it in millions.

Q12. Declare a boolean variable `isEven`. Assign it based on whether the number 42 is even using the modulus operator. Print the result.

● Operators (Q13–Q20)

Q13. Take two integers from the user and print the result of all 5 arithmetic operators (+, -, *, /, %).

Q14. Take a 3-digit number (e.g. 357). Use % and / operators to extract and print each digit separately.

Q15. Take marks of 5 subjects as variables. Calculate total, average, and print whether the student passed (average ≥ 40).

Q16. Write a program using only relational operators to check: is a equal to b? is a greater? is a less? Print all results.

Q17. A person can vote if age ≥ 18 AND isCitizen == true. Use logical operators to determine and print eligibility.

Q18. Use all shorthand assignment operators ($+=$, $-=$, $*=$, $/=$, $\%=$) on a starting value of 50 and print the result after each operation.

Q19. Print numbers 1 to 10 using only the increment operator ($++$) inside a while loop.

Q20. Take a number n. Print true if it is divisible by both 3 and 5, true if divisible by either, and true if not divisible by either. Use $\&\&$, $\|\|$, and $!$.

● Input / Output (Q21–Q26)

Q21. Accept a user's name and age via Scanner. Print: "Hello [name], you are [age] years old." using printf with formatting.

Q22. Accept two floating-point numbers from the user and print their sum, difference, product, and quotient — each on a new line using println.

Q23. Accept a temperature in Celsius from the user. Convert it to Fahrenheit using $F = (C * 9/5) + 32$ and print the result formatted to 2 decimal places.

Q24. Accept a character from the user using `sc.next().charAt(0)`. Print whether it is a vowel or a consonant using logical operators.

Q25. Accept the radius of a circle from the user. Calculate and print the area ($\pi * r^2$) and circumference ($2 * \pi * r$) using `Math.PI`.

Q26. Accept a 4-digit year from the user. Print whether it is a leap year. (Leap year: divisible by 4 AND (not divisible by 100 OR divisible by 400).)

● Mixed / Challenge (Q27–Q30)

Q27. Accept the length and width of a rectangle from the user. Print the perimeter, area, and whether it is a square (using a boolean).

Q28. Accept a student's name, roll number, and marks in 3 subjects. Calculate total, average, and assign a grade:

- $\geq 90 \rightarrow O, \geq 75 \rightarrow A, \geq 60 \rightarrow B, \geq 40 \rightarrow C, < 40 \rightarrow F$

Print a formatted result card.

Q29. Accept the principal, annual interest rate, and number of years. Calculate **Compound Interest** using $CI = P * (1 + R/100)^T - P$. Use `Math.pow()` and print the result to 2 decimal places.

Q30. Accept two integers a and b. Without using any if statement, print the maximum using this formula:

$$\text{max} = (a + b + \text{Math.abs}(a - b)) / 2$$

Print the minimum too using: $\text{min} = (a + b - \text{Math.abs}(a - b)) / 2$. Use `Math.abs()`.